Foundations of Data Science

Lecture 6: Markov chain Monte Carlo

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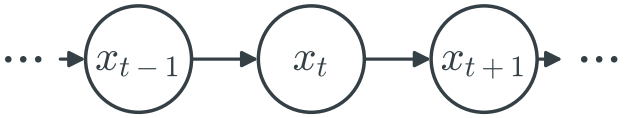


How to sample from a distribution $p(x)$ for which we can only evaluate an unnormalized version $\tilde{p}(x)$ of its density?

Use cases:

- Generating data points x from a generative model $p(x)$.
- Computing expectations $\mathbb{E}_{p(x)}[f(x)]$ or integrals $\int f(x)p(x)dx$.
- Computing posterior distributions $p(\mathbf{z} \mid \mathbf{x})$ in Bayesian inference.

Markov chain Monte Carlo



Markov chains

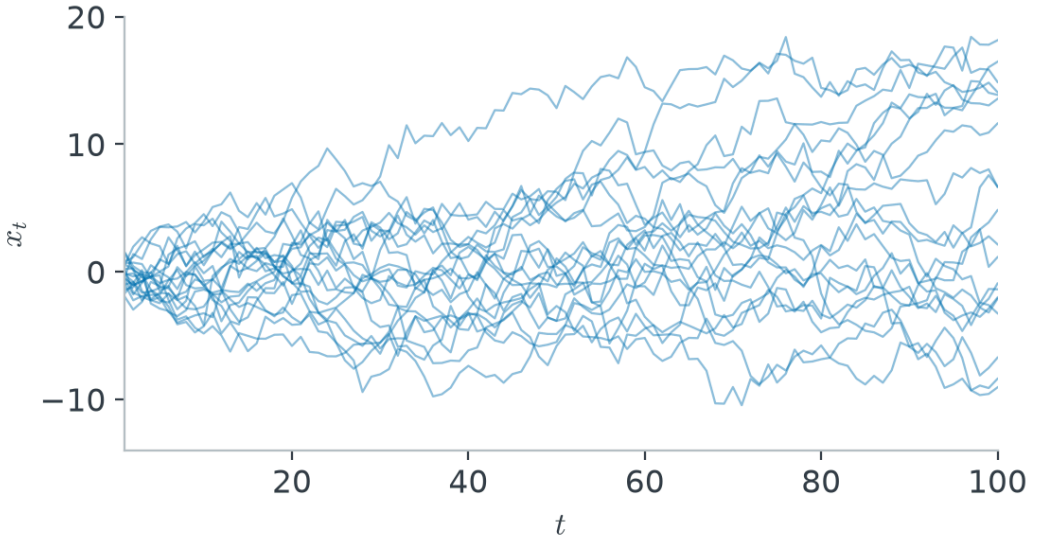
A Markov chain is a sequence of random variables $X_1, X_2, \ldots$ that assumes the conditional independence of X_{t+1} and $X_{1:t-1}$ given X_t.

It follows that the joint distribution of $X_1, \ldots, X_T$ can be factorized as

$$p(x_1, \ldots, x_T) = p(x_1) \prod_{t=1}^{T-1} p(x_{t+1} \mid x_t),$$

where $p(x_1)$ is the **initial distribution** and $p(x_{t+1} \mid x_t)$ is the **transition model**.

When the transition model does not depend on t, i.e., $p(x_{t+1} \mid x_t) = p(x' \mid x)$ for all t, the Markov chain is said to be **time-homogeneous**.



Example 1. Gaussian random walk with $p(x_1) = \mathcal{N}(x_1|0, 1)$ and transition model $p(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1}|x_t, \sigma^2)$.

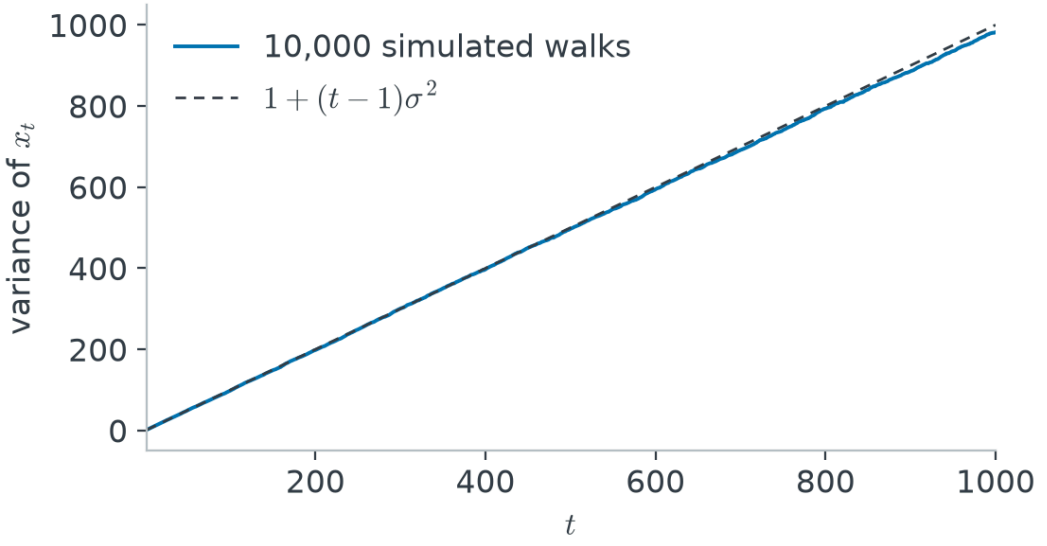
Stationary distributions

A stationary distribution π is a distribution that remains unchanged when passed through the transition model.

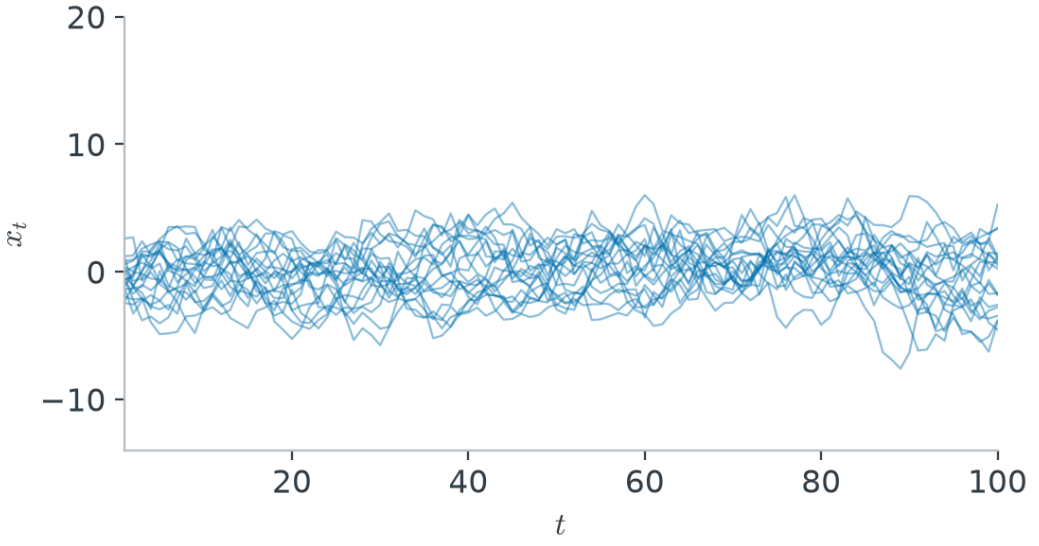
Formally, $\pi(x)$ is stationary for the transition model $p(x' \mid x)$ if

$$\pi(x') = \int \pi(x) p(x' \mid x) dx.$$

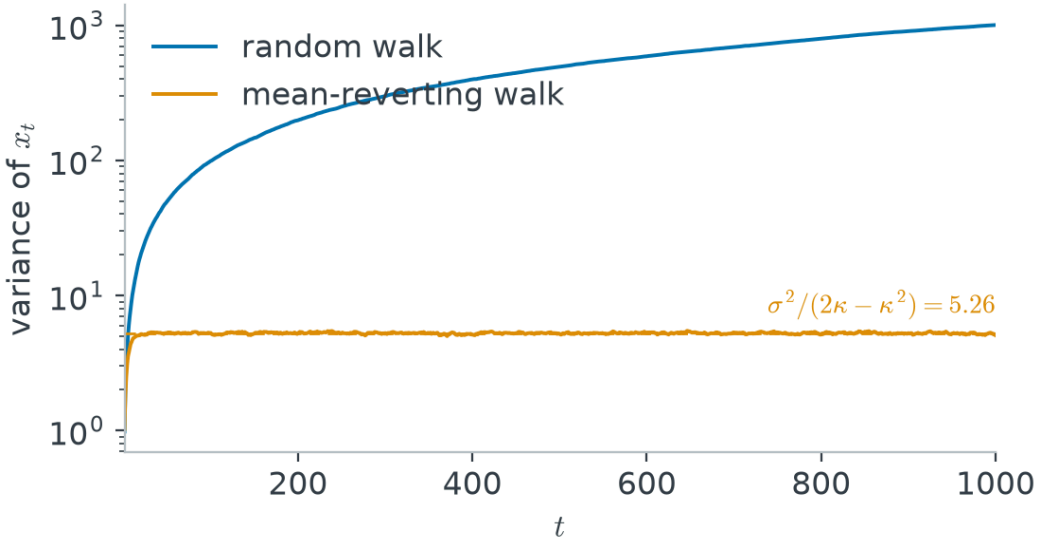
Therefore, if $x_t \sim \pi(x)$, then $x_{t+1} \sim \pi(x)$.



Example 1, continued. The Gaussian random walk does not have a stationary distribution since its variance increases indefinitely over time.



Example 2. Mean-reverting random walk with $p(x_1) = \mathcal{N}(x_1|0, 1)$ and transition model $p(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1}|x_t - \kappa(x_t - \mu), \sigma^2)$.



The mean-reverting random walk has a stationary distribution $\pi(x) = \mathcal{N}(x|\mu, \frac{\sigma^2}{2\kappa - \kappa^2})$.

Regularity conditions

A Markov chain is said to be:

- **irreducible** if any state x' can be reached from any state x, i.e., for any x, x', there exists t such that $p(x_t = x' \mid x_1 = x) > 0$.
- **aperiodic** if there is no deterministic cycle, i.e., for all x, the greatest common divisor of the set $\{t : p(x_t = x \mid x_1 = x) > 0\}$ is 1.
- for discrete state spaces, **positive recurrent** if the expected return time to any state is finite; for continuous state spaces, **positive Harris recurrent** if the chain returns to any set A with positive measure infinitely often with probability 1 and has a stationary distribution that integrates to one.

Basic limit theorem for Markov chains

Theorem. If a time-homogeneous Markov chain is irreducible, aperiodic, and positive (Harris) recurrent, then it satisfies the following properties:

- (Existence and uniqueness) There exists a **unique** stationary distribution π;
- (Convergence) For any initial distribution $p(x_1)$, $p(x_t) \to \pi(x)$ as $t \to \infty$;
- (Ergodic theorem) For any function f with $\mathbb{E}_\pi[|f(x)|] < \infty$,

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} f(x_t) = \mathbb{E}_\pi[f(x)]$$

  almost surely.

Proof. (Omitted and beyond scope.)

The theorem leaves two questions open: how to build a chain whose stationary distribution is a given target, and how fast the chain converges to it.

Detailed balance and Metropolis-Hastings answer the first. Diagnostics, at the end of the lecture, address the second.



Detailed balance

A transition model $p(x' \mid x)$ satisfies detailed balance with respect to a distribution π if

$$\pi(x)p(x' \mid x) = \pi(x')p(x \mid x')$$

for all x, x'.

Proposition. If a transition model $p(x' \mid x)$ satisfies detailed balance with respect to a distribution π, then π is a stationary distribution of $p(x' \mid x)$.

Proof. Integrating both sides of the detailed balance equation over x, we get

$$\begin{aligned}\int \pi(x)p(x' \mid x)dx &= \int \pi(x')p(x \mid x')dx \\ &= \pi(x') \int p(x \mid x')dx \\ &= \pi(x').\end{aligned}$$

Detailed balance is a **sufficient** condition for stationarity.

It is easier to check than stationarity itself, which makes it the tool of choice to construct a Markov chain with a given stationary distribution.

Metropolis-Hastings

The Metropolis-Hastings algorithm is a procedure for sampling from a target distribution $\pi(x)$, given only an unnormalized version $\tilde{\pi}(x)$ of its density.

The core idea is to construct a Markov chain whose stationary distribution matches the target distribution $\pi(x)$.

The Metropolis-Hastings algorithm proceeds as follows:

1. Given the current state x_t, propose a new state $x' \sim q(x' \mid x_t)$ using a proposal distribution $q(x' \mid x_t)$.
2. Compute the acceptance ratio

$$\alpha(x' \mid x_t) = \min\left(\frac{\tilde{\pi}(x')q(x_t \mid x')}{\tilde{\pi}(x_t)q(x' \mid x_t)}, 1\right).$$

3. Accept the proposed state x' with probability $\alpha(x' \mid x_t)$. If accepted, set $x_{t+1} = x'$. Otherwise, set $x_{t+1} = x_t$.
4. Repeat steps 1-3 for a large number of iterations.

This process generates a sequence of samples $x_1, x_2, \ldots$ that form a Markov chain.

Proposition. The Metropolis-Hastings algorithm constructs a Markov chain whose stationary distribution is $\pi(x)$.

Proof. The transition model of the Markov chain is given by

$$p(x' \mid x) = \begin{cases} q(x' \mid x)\alpha(x' \mid x) & \text{if } x' \neq x, \\ r(x) & \text{if } x' = x, \end{cases}$$

where $r(x) = 1 - \int q(x' \mid x)\alpha(x' \mid x)dx'$ is the probability of rejecting the proposal and staying at the current state.

To show that $\pi(x)$ is stationary, we verify the detailed balance condition for $x' \neq x$:

$$\begin{aligned}
\tilde{\pi}(x)p(x' \mid x) &= \tilde{\pi}(x)q(x' \mid x)\alpha(x' \mid x) \\
&= \tilde{\pi}(x)q(x' \mid x)\min\left(\frac{\tilde{\pi}(x')q(x \mid x')}{\tilde{\pi}(x)q(x' \mid x)}, 1\right) \\
&= \min(\tilde{\pi}(x)q(x' \mid x), \tilde{\pi}(x')q(x \mid x')) \\
&= \tilde{\pi}(x')q(x \mid x')\min\left(\frac{\tilde{\pi}(x)q(x' \mid x)}{\tilde{\pi}(x')q(x \mid x')}, 1\right) \\
&= \tilde{\pi}(x')p(x \mid x').
\end{aligned}$$

Since $\pi(x) = \frac{\tilde{\pi}(x)}{Z}$ for some normalization constant Z, the detailed balance condition also holds for $\pi(x)$. Hence, $\pi(x)$ is a stationary distribution of the Markov chain.

For $x' = x$, detailed balance holds trivially.

Proposition. The Markov chain constructed by the Metropolis-Hastings algorithm satisfies the regularity conditions under appropriate conditions on q.

Sketch of proof.

- Aperiodicity: The chain can stay at the same state when proposals are rejected, so $p(x|x) > 0$. This breaks any periodic cycles.
- Irreducibility: If the proposal q has appropriate support, then for any x, x' where $\pi(x), \pi(x') > 0$, there exists a sequence of proposals with positive probability connecting them.
- Recurrence: If π is a proper probability distribution, the chain is positive Harris recurrent. (Proof beyond scope)

These conditions ensure convergence to $\pi(x)$ by the basic limit theorem.

Proposal distributions

The efficiency of the Metropolis-Hastings algorithm heavily depends on the choice of the proposal distribution $q(x' \mid x)$.

Common choices include:

- **Random walk**: $q(x' \mid x) = \mathcal{N}(x'|x, \sigma^2 I)$ for continuous state spaces. Simple and general, but efficiency depends on tuning σ (optimal acceptance rate: about 44% in one dimension, 23% in high dimension).
- **Langevin**: $q(x' \mid x) = \mathcal{N}(x'|x + \frac{\epsilon^2}{2}\nabla \log \tilde{\pi}(x), \epsilon^2 I)$, which uses gradient information to guide proposals toward high-probability regions. Requires differentiable $\tilde{\pi}(x)$.
- **Independence**: $q(x' \mid x) = q(x')$, independent of current state. Can be efficient if $q(x')$ closely approximates $\pi(x)$.

Hamiltonian Monte Carlo

Random-walk proposals explore the target by small, undirected steps. **Hamiltonian Monte Carlo** (HMC) proposes distant states by following the gradient of $\log \tilde{\pi}$.

The state $x \in \mathbb{R}^d$ is augmented with a momentum $p \in \mathbb{R}^d$, and the chain targets the joint distribution

$$\pi(x, p) \propto \tilde{\pi}(x)\mathcal{N}(p|0, I) \propto \exp(-H(x, p))$$

defined by the **Hamiltonian**

$$H(x, p) = -\log \tilde{\pi}(x) + \frac{1}{2}p^T p.$$

Hamiltonian dynamics

The Hamiltonian dynamics

$$\frac{dx}{ds} = \frac{\partial H}{\partial p} = p, \quad \frac{dp}{ds} = -\frac{\partial H}{\partial x} = \nabla \log \tilde{\pi}(x)$$

conserve H, preserve volume and are reversible. A proposal made by following them would be accepted with probability 1.

The dynamics are integrated numerically by the **leapfrog** scheme, which repeats L steps of size ϵ:

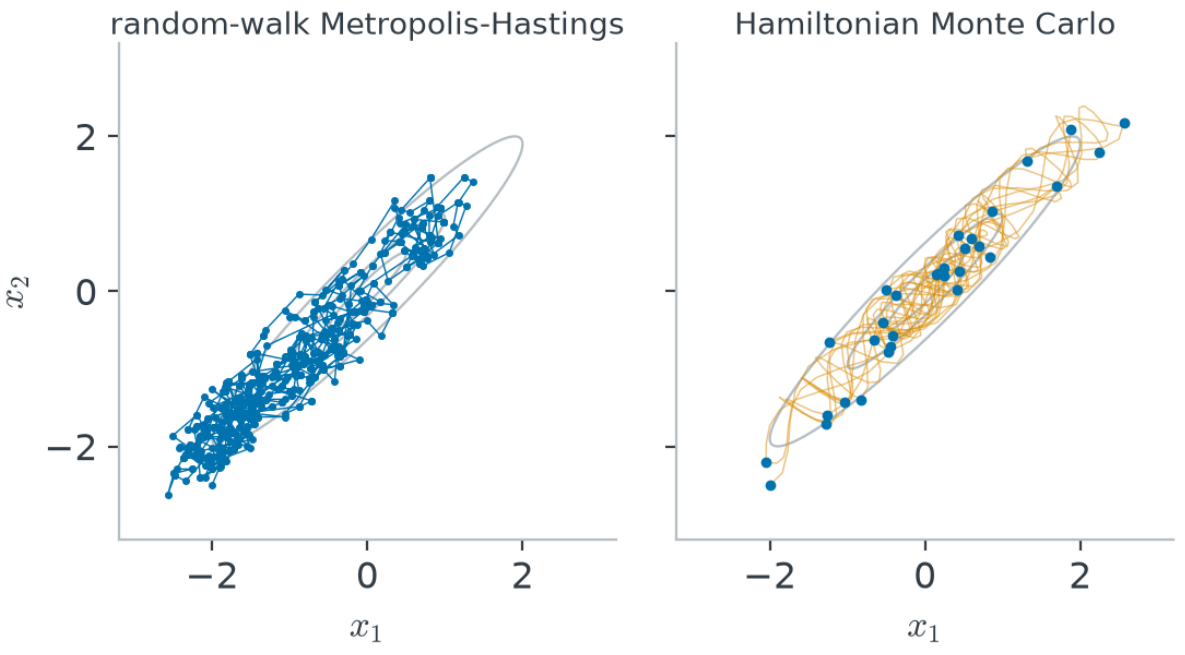
$$\begin{aligned} p &\leftarrow p + \frac{\epsilon}{2} \nabla \log \tilde{\pi}(x) \\ x &\leftarrow x + \epsilon p \\ p &\leftarrow p + \frac{\epsilon}{2} \nabla \log \tilde{\pi}(x). \end{aligned}$$

The HMC algorithm proceeds as follows:

1. Given the current state x_t, draw a momentum $p \sim \mathcal{N}(0, I)$.
2. Starting from (x_t, p), run L leapfrog steps of size ϵ to obtain (x', p').
3. Accept $x_{t+1} = x'$ with probability

$$\alpha = \min\left(1, \exp(H(x_t, p) - H(x', p'))\right).$$

   Otherwise, set $x_{t+1} = x_t$.



Random-walk Metropolis-Hastings (600 iterations) and HMC (30 iterations of 20 leapfrog steps, in orange) on a correlated Gaussian, for the same number of evaluations of the target.

Other samplers

- **No-U-Turn Sampler (NUTS)**: An adaptive variant of HMC that automatically tunes the trajectory length L. Default sampler in Stan and PyMC.
- **Gibbs sampling**: Updates variables one at a time from conditional distributions. A special case of MH with acceptance probability = 1 when conditionals are tractable.



Run the first part of `nb06b-mcmc.ipynb` to see Metropolis-Hastings in action.

Bayesian inference with MCMC

Bayesian inference with MCMC

In Bayesian inference, we are often interested in computing the posterior distribution

$$p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}}) = \frac{p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta)}{p(\mathbf{x}_{\text{obs}})}$$

of latent variables $\mathbf{z}$ and parameters θ given observed data $\mathbf{x}_{\text{obs}}$.

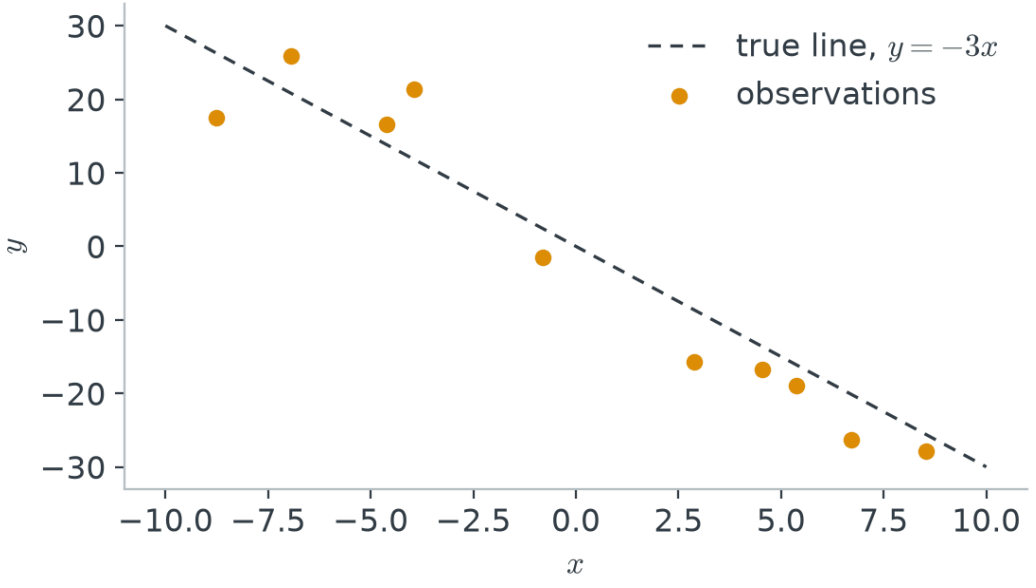
> In general, the normalizer $p(\mathbf{x}\text{obs}) = \int p(\mathbf{x}\text{obs} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta) d\mathbf{z} d\theta$ has no closed form.

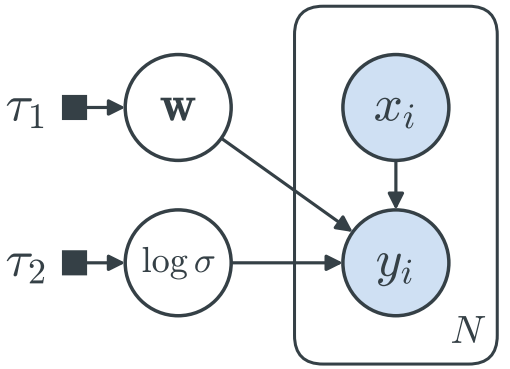


MCMC methods, such as Metropolis-Hastings, generate samples from the posterior $p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}})$ even when the normalizer $p(\mathbf{x}_{\text{obs}})$ has no closed form.

Since MCMC methods only require an unnormalized density $\tilde{p}$, **the joint density $p(\mathbf{x}_{\text{obs}}, \mathbf{z}, \theta) = p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta)\, p(\mathbf{z}, \theta)$ can be used as the target**, without the normalizer $p(\mathbf{x}_{\text{obs}})$.

Example 3. Bayesian linear regression. Consider a data set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ of N pairs of an input $x_i \in \mathbb{R}$ and a target $y_i \in \mathbb{R}$.





We model y given x as a line with Gaussian noise,

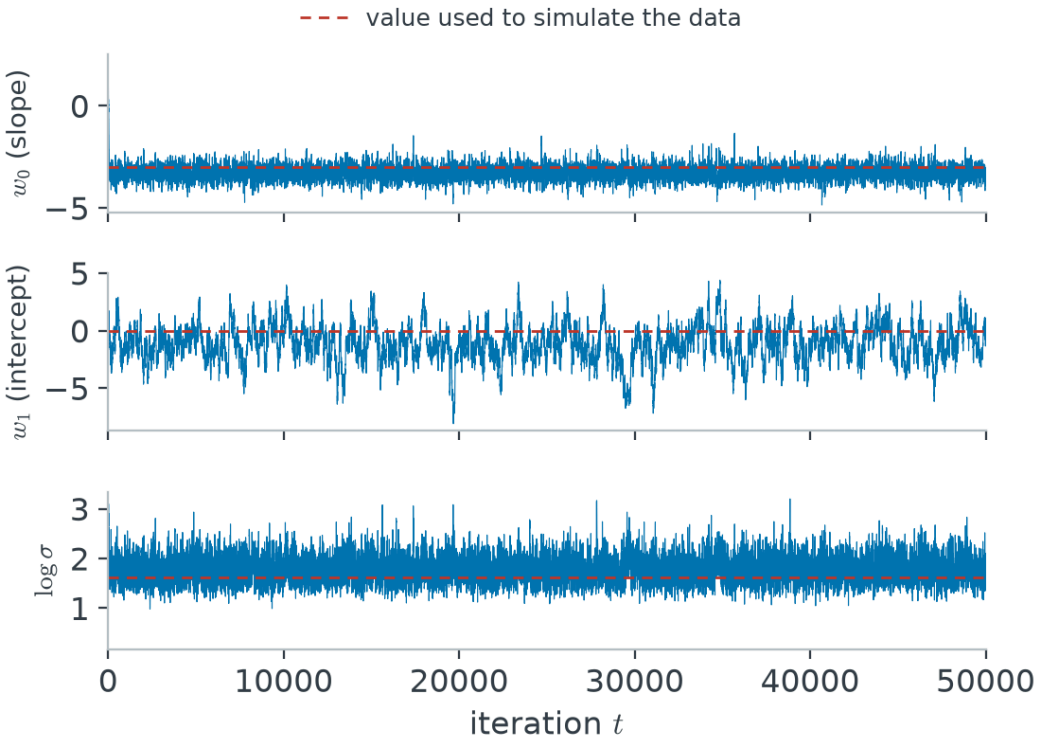
$$p(y \mid x, \mathbf{w}, \log \sigma) = \mathcal{N}(y \mid w_0 x + w_1, \sigma^2),$$

where $\mathbf{w} = (w_0, w_1)$ holds the slope and the intercept. The priors are $p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \tau_1^2 \mathbf{I})$ and $p(\log \sigma) = \mathcal{N}(\log \sigma \mid 0, \tau_2^2)$, with hyperparameters τ_1 and τ_2.

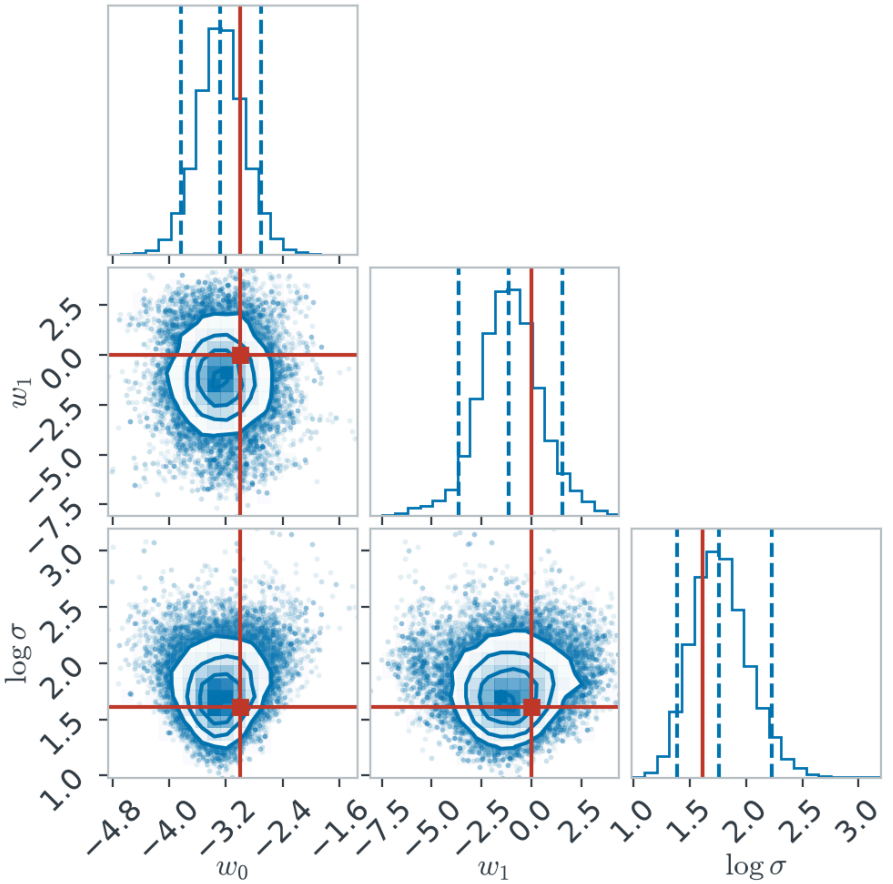
The target distribution is the posterior $p(\mathbf{w}, \log \sigma \mid \mathcal{D})$, which is proportional to the joint density

$$p(\mathbf{w}, \log \sigma \mid \mathcal{D}) \propto \left(\prod_{i=1}^{N} p(y_i \mid x_i, \mathbf{w}, \log \sigma) \right) p(\mathbf{w}) p(\log \sigma).$$

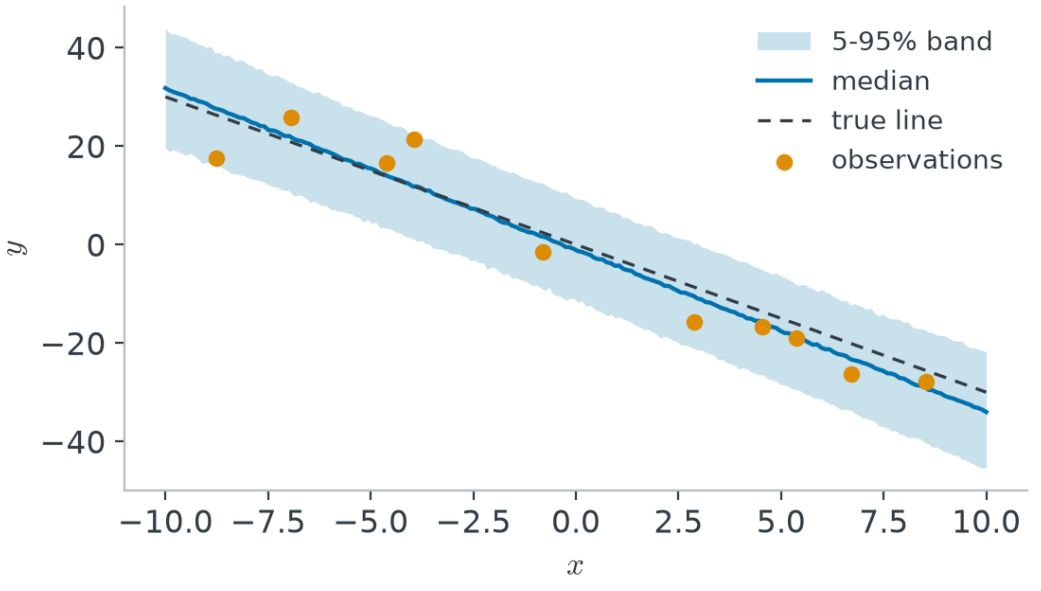
We can use Metropolis-Hastings to sample from this posterior distribution using the joint density as the unnormalized target density.



Traces of w_0, w_1 and $\log \sigma$ along the Markov chain of Metropolis-Hastings.



Posterior samples, after a warm-up of 1000 iterations.



Posterior predictive distribution $p(y \mid x, \mathcal{D})$: median and 5-95% band.

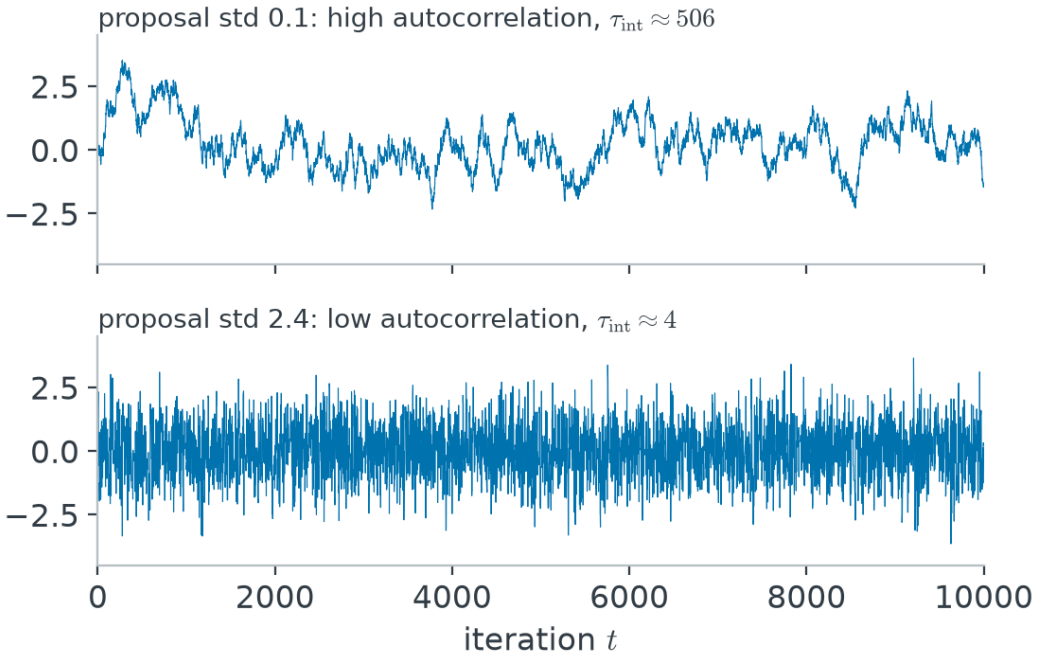


Diagnostics for MCMC

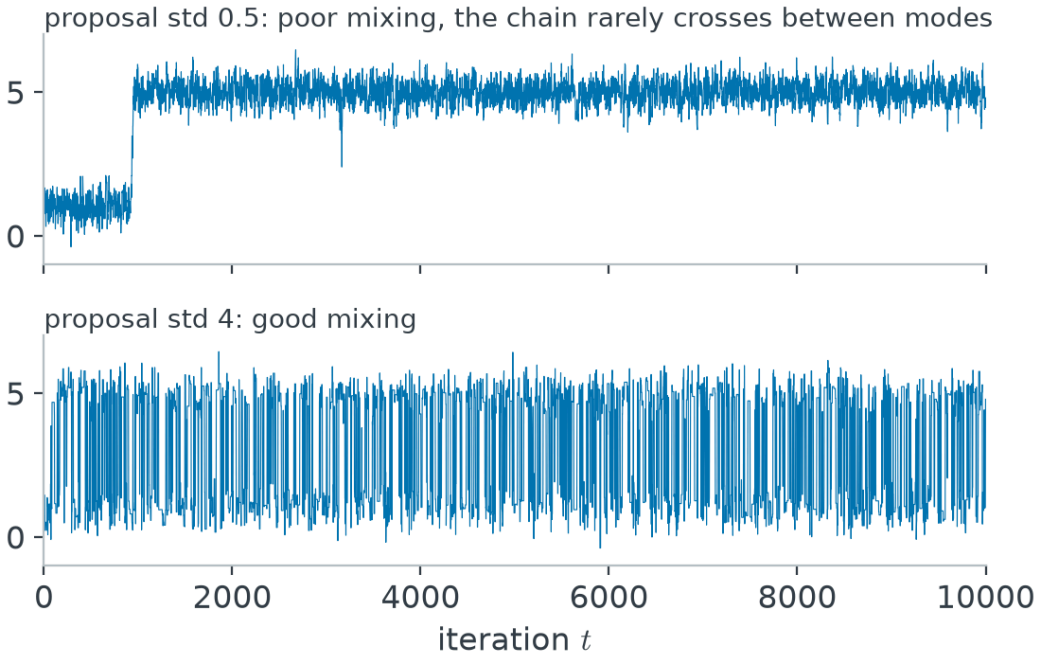
MCMC algorithms produce only approximate samples from the target distribution. It is important to assess the quality of these samples using diagnostics in terms of **autocorrelation**, **mixing** and **convergence**.

Trace plots of the Markov chains can reveal mixing, autocorrelation, and convergence issues:

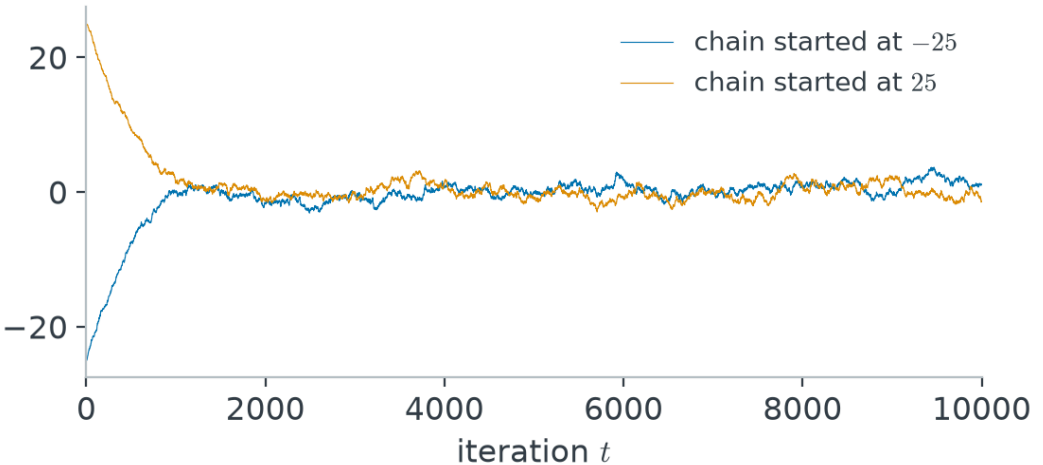
- High autocorrelation is indicated by chains that exhibit strong correlations between successive samples.
- Poor mixing is indicated by chains that explore the state space slowly or get stuck in certain regions.
- When running multiple chains, lack of convergence is indicated by chains that do not overlap or converge to the same distribution.



High vs. low autocorrelation. For highly autocorrelated chains, mixing is poor and effective sample size is low.



Poor vs. good mixing for a multimodal target. Poor mixing can be due to small proposal steps or energy barriers between modes.



Two chains run on the same target but initialized differently.
Traces show convergence to the same distribution.

Acceptance rate

The first quantitative diagnostic to monitor is the **acceptance rate**, defined as the fraction of proposed samples that are accepted.

- A very low acceptance rate (e.g., < 10%) indicates that the proposal distribution is making too large jumps, leading to many rejections.
- A very high acceptance rate (e.g., > 80%) suggests that the proposal distribution is making too small jumps, resulting in slow exploration of the state space.
- For a Gaussian random-walk proposal, the optimal acceptance rate is about 44% in one dimension and decreases to 23.4% as the dimension grows.

Effective Sample Size

Quantitatively, the **effective sample size**

$$\mathrm{ESS} = \frac{T}{1 + 2\sum_{k=1}^{\infty} \rho_k}$$

accounts for autocorrelation in MCMC samples, where T is the number of samples in the chain and

$$\rho_k = \frac{\mathrm{Cov}(X_t, X_{t+k})}{\mathrm{Var}(X_t)}$$

is the **autocorrelation at lag** k computed over the samples.

A higher ESS indicates more independent samples and better mixing.

Integrated Autocorrelation Time

The **integrated autocorrelation time**

$$\tau_{\text{int}} = 1 + 2\sum_{k=1}^{\infty} \rho_k$$

quantifies the number of correlated samples equivalent to one independent sample.

A smaller τ_{int} indicates better mixing and more efficient sampling.

The effective sample size is therefore $\text{ESS} = T/\tau_{\text{int}}$.

Gelman-Rubin Statistic

The **Gelman-Rubin statistic** $\hat{R}$ serves as a convergence diagnostic when running M chains in parallel, each with T samples (after burn-in). Let

$$W = \frac{1}{M} \sum_{m=1}^{M} s_m^2$$

be the within-chain variance, where s_m^2 is the sample variance of chain m, and

$$B = \frac{T}{M-1} \sum_{m=1}^{M} (\bar{x}_m - \bar{x})^2$$

be the between-chain variance, where $\bar{x}_m$ is the sample mean of chain m and $\bar{x}$ is the overall mean across chains. Then,

$$\hat{R} = \sqrt{\frac{\frac{T-1}{T} W + \frac{1}{T} B}{W}}.$$

Rank-normalized split-$\hat{R}$

In practice, $\hat{R}$ is computed with two modifications:

- each chain is split into two halves, which detects a chain that drifts during the run;
- each draw is replaced by the normal quantile of its rank among all draws, which makes $\hat{R}$ robust to heavy tails.

Values of $\hat{R}$ below 1.01 indicate convergence across chains.

The effective sample size is computed on the same rank-normalized draws (**bulk ESS**) and on the indicators of the 5% and 95% quantiles (**tail ESS**). Both should exceed 100 per chain.

Calibration

Diagnostics check that the chains have converged. They do not check that they converged to the right posterior: a bug in the log density goes unnoticed.

Let $\theta^* \sim p(\theta)$ and $\mathcal{D} \sim p(\mathcal{D} \mid \theta^*)$. The exact posterior is **calibrated**: a credible region $C_\alpha(\mathcal{D})$ of level α contains θ^* with probability α,

$$P(\theta^* \in C_\alpha(\mathcal{D})) = \alpha.$$

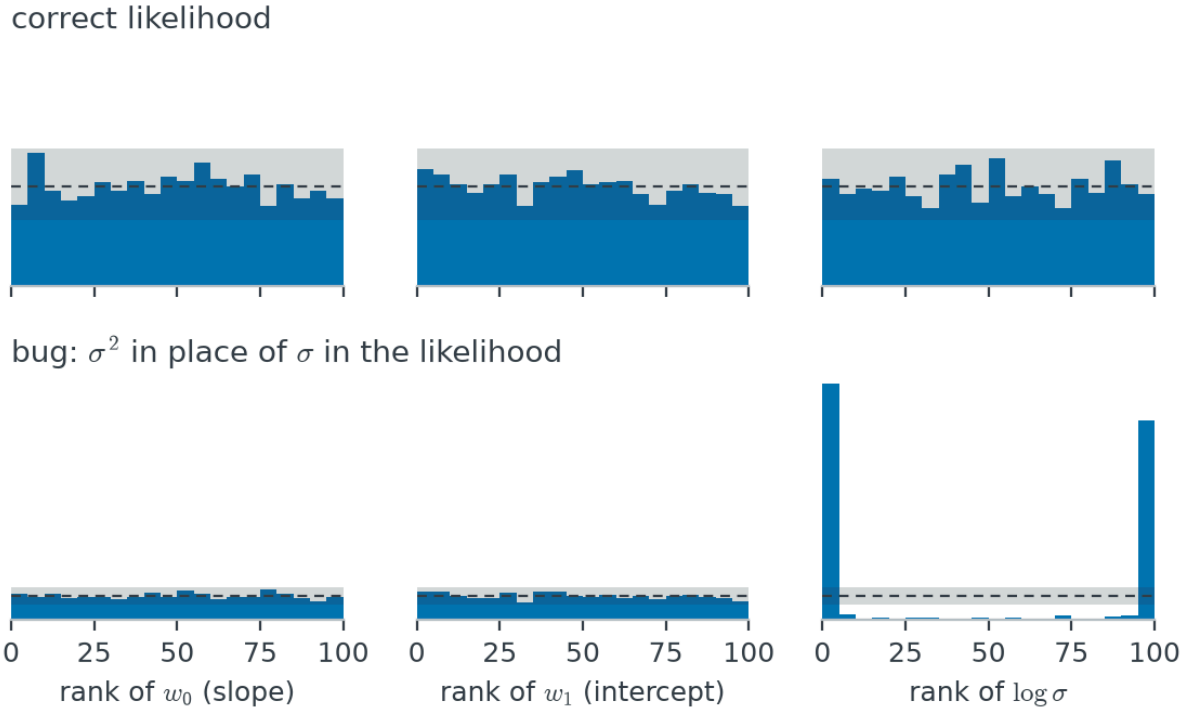
Simulation-based calibration

For $s = 1, \ldots, S$:

1. Draw $\theta_s^* \sim p(\theta)$ and $\mathcal{D}_s \sim p(\mathcal{D} \mid \theta_s^*)$.
2. Draw L samples $\theta_1, \ldots, \theta_L$ with the sampler targeting $p(\theta \mid \mathcal{D}_s)$.
3. Compute the rank $r_s = \sum_{l=1}^{L} 1[\theta_l < \theta_s^*] \in \{0, \ldots, L\}$.

Proposition. If the sampler draws independent samples from the exact posterior, the ranks r_s are uniformly distributed over $\{0, \ldots, L\}$.

Proof. Given $\mathcal{D}_s$, $\theta_s^*, \theta_1, \ldots, \theta_L$ are $L + 1$ independent draws from $p(\theta \mid \mathcal{D}_s)$. By symmetry, θ_s^* takes any of the $L + 1$ positions in their sorted order with the same probability.



Rank histograms of simulation-based calibration for the Bayesian linear regression, with $S = 1000$ simulations and $L = 99$ draws of random-walk Metropolis-Hastings each. The grey band holds 99% of the counts of uniform ranks.

The end.